

Two Measurements per Technician

AP Statistics · Unit 4 · Topic 4.4 · B: 2027-01-19 · E: 2027-03-05 · TEACHER KEY

CED TETHER

- **Skill 1.E** — Identify an appropriate inference method for significance tests.
- **Skill 1.F** — Identify null and alternative hypotheses.
- **Skill 4.C** — Verify that inference procedures apply in a given situation.
- **EU VAR-7** — The t-distribution may be used to model variation.
- **LO VAR-7.B** — Identify an appropriate testing method for a population mean with unknown σ , including the mean difference between values in matched pairs. [Skill 1.E]
- **EK VAR-7.B.1** — The appropriate test for a population mean with unknown σ is a one-sample t-test for a population mean.
- **EK VAR-7.B.2** — Matched pairs can be thought of as one sample of pairs. Once differences between pairs of values are found, inference for significance testing proceeds as for a population mean.
- **LO VAR-7.C** — Identify the null and alternative hypotheses for a population mean with unknown σ , including the mean difference between values in matched pairs. [Skill 1.F]
- **EK VAR-7.C.1** — The null hypothesis for a one-sample t-test for a population mean is $H_0: \mu = \mu_0$, where μ_0 is the hypothesized value. Depending upon the situation, the alternative hypothesis is $H_a: \mu < \mu_0$, or $H_a: \mu > \mu_0$, or $H_a: \mu \neq \mu_0$.
- **EK VAR-7.C.2** — When finding the mean difference, μ_D , between values in a matched pair, it is important to define the order of subtraction.
- **LO VAR-7.D** — Verify the conditions for the test for a population mean, including the mean difference between values in matched pairs. [Skill 4.C]
- **EK VAR-7.D.1** — In order to make statistical inferences when testing a population mean, we must check for independence and that the sampling distribution is approximately normal:
 - a. To check for independence:
 - (i) Data should be collected using a random sample or a randomized experiment.
 - (ii) When sampling without replacement, check that $n \leq 10\% N$.
 - b. To check that the sampling distribution of $\bar{x}$ is approximately normal (shape):
 - (i) If the observed distribution is skewed, n should be greater than 30.
 - (ii) If the sample size is less than 30, the distribution of the sample data should be free from strong skewness and outliers.

Essential question: How does the study structure determine the parameter, hypotheses, and conditions for a mean test?

DOK-3 skill: 4.C · **FRQ pattern:** paired-test-setup-and-consequences

DOK rationale: Part (c) coordinates repeated measures, subtraction order, tail direction, independent sample size, and shape evidence, then traces the uncertainty and research-question consequences of the rejected setup.

Do Now (first take) → Explore (video + follow-along) → Exit (finish + turn in) DOK: 1
→ 3 **Minutes: 45**

Teacher does

- Display the board and ask for one sentence using the study description.
- Begin the 7.4 video follow-along at minute 5.
- Return to the board at minute 33. Students finish the shortened parts and justify their judgment.

- Collect at the door. Score part (c) only using the three required elements.

Students do

- Write a one-sentence first take before the video.
- Complete the video follow-along worksheet.
- Finish (a)–(c), revise the first take in the exit line, and turn in the sheet.

Questions to ask

- Which specific number or study detail supports your choice?
- What claim would go beyond the evidence, and why?

Adult role

Circulate and ask for evidence. Accept a concise response; do not require omitted calculations or a second full inference write-up.

[WARN] WATCH FOR

- Choosing a speaker without a reason.
- Copying numbers without explaining what they support.

SCORING PART (c) — E / P / I

Expected elements

- **paired-units** (required) — Justifies a one-sample test of the 18 within-technician differences
- **correct-hypotheses** (required) — Uses the population mean difference, zero null, and lower-tail alternative
- **rejects-independent-times** (required) — Explains that the two times from each person are linked

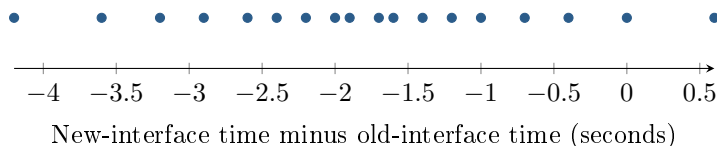
E	Justifies the paired units and explains the lower-tail population hypotheses using new minus old and the reduction question.
P	Shows substantive valid reasoning for at least one required element, but does not meet all E requirements. Credit correct reasoning even when another claim is wrong.
I	Shows no substantive valid reasoning for any required element; a name, unsupported choice, or copied number alone does not count.

[STAR] TODAY'S PROBLEM — annotated

Suppose a company takes an SRS of 18 of its 300 technicians. Each completes the same task with old and new interfaces; interface order is randomly assigned. The plot shows $d = \text{new time} - \text{old time}$. Researchers ask whether the new interface reduces the population mean time.

Nia: “The technician seems to be the independent unit, so I would build one difference per person; the subtraction order matters for the tail.”

Colin: “There are 36 recorded times, so I would use two independent samples of 18 and a two-sided alternative.”



FIRST TAKE (5 min)

Would you compare each technician's two times or treat them as separate people? Explain in one sentence.

Key: Ungraded. Everyday language is enough before the video; formal vocabulary belongs in the finish.

Rules callout “SET UP A TEST FOR A MEAN (keep on the page)” is printed on the student sheet.

DOK 1 (a) Define μ_d in context.

Key: μ_d is the mean new-minus-old task time for all 300 technicians.

DOK 2 (b) Check the random-sample, 10 percent, and small-sample shape conditions in three short notes.

Key: The 18 technicians were randomly sampled. $18 \leq 0.10(300) = 30$. Since $n = 18 < 30$, inspect the differences: the plot shows no strong skewness or outliers.

DOK 3 (c) In 3–4 sentences, choose a setup and justify it. State the hypotheses using the given subtraction order and reduction question, then explain why the 36 times are not 36 independent people.

Key: Nia is justified: use a matched-pairs one-sample t -test on the 18 differences. For new minus old, a reduction is negative, so $H_0 : \mu_d = 0$ and $H_a : \mu_d < 0$. Each technician supplies two linked times, so there are 18 independent people, not 36. Treating the times as independent loses the pairing, and a two-sided alternative asks about any change rather than the planned reduction.

EXIT

One thing the video changed about my first take: _____